1. Admin Module

Admin should be the central control panel.

Admin Login

- Username / Email
- Password
- Forgot Password
- OTP / 2FA
- Role-based access

Admin Roles

Role	Access
Super Admin	Full system
Transport Manager	Vehicles, drivers, routes, trips
Operations	Trip allocation & monitoring
HR	Employee details
Finance	Fuel, billing & expenses
Viewer	Reports only

Admin Dashboard

- Total Vehicles
- Active Vehicles
- Vehicles Under Maintenance
- Total Drivers
- Available Drivers
- Today's Trips
- Completed Trips
- Cancelled Trips
- Employees Transported
- Today's Fuel Consumption
- Fuel Cost
- Vehicle Utilization
- Driver Performance

2. Vehicle Master

Vehicle Registration

```
Vehicle Number
       ↓
Vehicle Type
       ↓
Make / Model
       ↓
Fuel Type
       ↓
Seating Capacity
       ↓
Year of Manufacture
       ↓
Insurance Details
       ↓
FC Details
       ↓
PUC Details
       ↓
Permit Details
       ↓
GPS Device
       ↓
Status
```

Vehicle Fields

- Vehicle ID
- Registration Number
- Vehicle Type
  - Sedan
  - SUV
  - Hatchback
  - Tempo Traveller
  - Bus
  - EV
- Make
- Model
- Fuel Type

    - Petrol
    - Diesel
    - CNG
    - EV
- Seating Capacity
- Chassis Number
- Engine Number
- Purchase Date
- Owner/Vendor
- Insurance Expiry
- FC Expiry
- PUC Expiry
- Permit Expiry
- GPS ID
- Current Odometer
- Status

Vehicle Status

```
AVAILABLE
   │
   ├── ASSIGNED
   │
   ├── ON TRIP
   │
   ├── MAINTENANCE
   │
   └── INACTIVE
```

3. Driver Master

Maintain complete driver records.

Driver Registration

```
Driver Details
      ↓
License Details
      ↓
Documents
      ↓
Medical / Verification
      ↓
Training
      ↓
Vehicle Allocation
      ↓
Driver Status
```

Driver Fields

- Driver ID
- Driver Name
- Mobile Number
- Date of Birth
- Address
- Emergency Contact
- Driving License Number
- License Type
- License Expiry
- Badge Number
- Aadhaar/ID reference
- Police Verification
- Medical Certificate
- Joining Date
- Vendor
- Experience
- Assigned Vehicle
- Status

Driver Status

```
AVAILABLE
ASSIGNED
ON TRIP
LEAVE
SUSPENDED
INACTIVE
```

---

4. Employee Master

Employees who use transportation.

Employee Details

- Employee ID
- Employee Name
- Mobile
- Email
- Department
- Designation
- Shift
- Pickup Location
- Drop Location
- Emergency Contact
- Gender
- Transport Required
- Transport Type
- Status

Employee Flow

```
Employee Registration
        ↓
Pickup Location
        ↓
Shift Selection
        ↓
Route Allocation
        ↓
Vehicle Allocation
        ↓
Driver Allocation
        ↓
Trip
```

5. Route Master

Very important for employee transportation.

Route Creation

```
Route Name
    ↓
Shift
    ↓
Pickup Points
    ↓
Employee Mapping
    ↓
Vehicle Capacity
    ↓
Driver
    ↓
Route Activation
```

Example:

Route: Chennai North – Shift A

```
Pickup 1 → Pickup 2 → Pickup 3 → Office
   5          7          4
 employees employees employees
```

System should automatically calculate:

- Total employees
- Distance
- Estimated travel time
- Vehicle capacity
- Route utilization

---

6. Vehicle + Driver Allocation

Admin selects:

```
DATE
 ↓
SHIFT
 ↓
ROUTE
```

```
 ↓
VEHICLE
 ↓
DRIVER
 ↓
EMPLOYEES
 ↓
CONFIRM TRIP
```

System should prevent:

- Same vehicle assigned to two trips at same time
- Same driver assigned to overlapping trips
- Vehicle capacity exceeded
- Expired vehicle documents
- Expired driver license

---

7. Trip Management

This is the core operational module.

Trip Lifecycle

```
TRIP CREATED
     ↓
ASSIGNED
     ↓
DRIVER ACCEPTED
     ↓
VEHICLE READY
     ↓
TRIP STARTED
     ↓
EMPLOYEE PICKUP
     ↓
EN ROUTE
     ↓
OFFICE ARRIVAL
     ↓
TRIP COMPLETED
```

Trip Screen

Admin should see:

- Trip ID
- Date
- Shift
- Route
- Vehicle
- Driver
- Employee count
- Pickup points
- Start time
- End time
- Start KM
- End KM
- Total KM
- Trip status
- GPS location

8. Employee Pickup / Attendance

Driver app can show:

```
Trip Started
     ↓
Pickup Point 1
     ↓
Employee Arrived?
 YES / NO
     ↓
OTP / QR / Attendance
     ↓
Pickup Confirmed
     ↓
Next Pickup
```

Employee status:

- Scheduled
- Waiting
- Boarded
- No Show

- Cancelled
- Dropped

---

9. Fuel Flow Module

This is one of the most important modules for fleet control.

Fuel Entry

```
Vehicle
   ↓
Fuel Date
   ↓
Fuel Type
   ↓
Opening KM
   ↓
Fuel Quantity
   ↓
Fuel Rate
   ↓
Total Amount
   ↓
Closing KM
   ↓
Fuel Station
   ↓
Receipt Upload
```

Automatically Calculate

Fuel Cost

`Fuel Quantity × Fuel Rate`

KM Run

`Closing KM − Opening KM`

Mileage

`KM Run ÷ Fuel Quantity`

Example:

```
Opening KM       25,000
Closing KM       25,450
KM Run              450 KM

Fuel               45 Litres
Fuel Cost          ₹4,500

Mileage             10 KM/L
Cost/KM             ₹10
```

Fuel Alerts

System should flag:

- Abnormal fuel consumption
- Low mileage
- Duplicate fuel entry
- Fuel entered without trip
- Fuel quantity exceeding tank capacity
- Sudden increase in fuel cost

---

10. Vehicle Maintenance

Maintenance Flow

```
Vehicle
   ↓
Service Due
   ↓
Service Request
   ↓
Garage / Vendor
   ↓
Work Order
   ↓
Parts + Labour
   ↓
Invoice
   ↓
Payment
   ↓
Vehicle Released
```

Track:

- Service date
- Odometer
- Service type
- Parts replaced
- Cost
- Next service KM
- Next service date

11. Document Expiry Management

The system should automatically monitor:

- Insurance
- FC
- PUC
- Permit
- Driver License
- Driver Badge
- Medical Certificate

12. GPS / Live Tracking

If GPS integration is available:

```
Vehicle
   ↓
GPS Device
   ↓
Live Location
   ↓
API
   ↓
Transportation App
   ↓
Admin Dashboard
```

Admin can view:

- Live vehicle location
- Speed
- Trip status
- Route deviation

- Idle time
- Distance
- ETA

13. Alerts & Notifications

Admin Notifications

- Trip started
- Trip delayed
- Vehicle breakdown
- Driver absent
- Employee no-show
- Route deviation
- Fuel entry
- Document expiry
- Maintenance due

Employee Notifications

- Trip assigned
- Driver details
- Vehicle number
- Driver arriving
- Pickup reminder
- Trip started
- Trip completed
- ETA

14. Reports

Vehicle Reports

- Vehicle utilization
- Vehicle-wise KM
- Vehicle-wise fuel
- Fuel cost/KM
- Maintenance cost
- Vehicle profitability

Driver Reports

- Trips completed
- Attendance
- Late trips
- Complaints
- KM driven
- Driver performance

Employee Reports

- Employee transport usage
- No-show report
- Pickup/drop report
- Shift-wise usage

Financial Reports

- Fuel expense
- Maintenance expense
- Driver cost
- Vehicle cost
- Route cost
- Cost per employee
- Cost per KM